1) $A = \iint_R \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\, dA$

$R = [0,5] \times [1,4]$

$z = 2 + 3x + 4y$

$= \int_1^4 \int_0^5 \sqrt{1 + 3^2 + 4^2}\, dx\, dy$

$= \int_1^4 \int_0^5 \sqrt{26}\, dx\, dy = 15\sqrt{26}$.

2) $A = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\, dA$

$D = \{(x,y) : x^2 + y^2 \le 1\}$

$z = xy$

$= \int_0^{2\pi} \int_0^1 \sqrt{1 + y^2 + x^2}\, r\, dr\, d\theta$

$= \int_0^{2\pi} \int_0^1 \sqrt{1 + r^2}\, r\, dr\, d\theta$

$u = 1 + r^2$

$du = 2r\, dr$

$= \int_0^{2\pi} \int_1^2 \frac{1}{2}\sqrt{u}\, du\, d\theta$

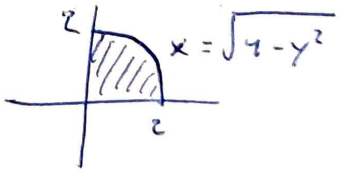
$= \int_0^{2\pi} \left[\frac{1}{3} u^{\frac{3}{2}}\right]_{u=1}^{u=2} d\theta$

$= \int_0^{2\pi} \frac{2}{3}\sqrt{2} - \frac{1}{3}\, d\theta = \frac{2\pi}{3}\left(2\sqrt{2} - 1\right)$.

3) $\iiint_E 2x\, dV$

$E = \{(x,y,z) : y \in [0,2]$
$x \in [0, \sqrt{4-y^2}]$
$z \in [x, 2x]\}$

$= \int_0^2 \int_0^{\sqrt{4-y^2}} \int_x^{2x} 2x\, dz\, dx\, dy$



$= \int_0^2 \int_0^{\sqrt{4-y^2}} 2x^2\, dx\, dy$

$= \int_0^{\pi/2} \int_0^2 2r^3 \cos^2\theta\, dr\, d\theta$

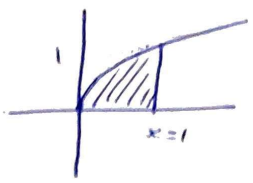
$= \int_0^{\pi/2} \left[\frac{1}{2} r^4 \cos^2\theta\right]_{r=0}^{r=2} d\theta$

$\cos^2\theta = \frac{1 + \cos 2\theta}{2}$

$= \int_0^{\pi/2} 8\cos^2\theta\, d\theta$

$= 4\int_0^{\pi/2} 1 + \cos 2\theta\, d\theta = 4\left[\theta + \frac{1}{2}\sin 2\theta\right]_0^{\pi/2} = 2\pi$.

4 $\iiint_E 6xy\,dV$ $\quad 0 \le z \le 1+x+y$



$$= \int_0^1 \int_0^{\sqrt{x}} \int_0^{1+x+y} 6xy\,dz\,dy\,dx$$

$$= \int_0^1 \int_0^{\sqrt{x}} 6xy + 6x^2y + 6xy^2\,dy\,dx$$

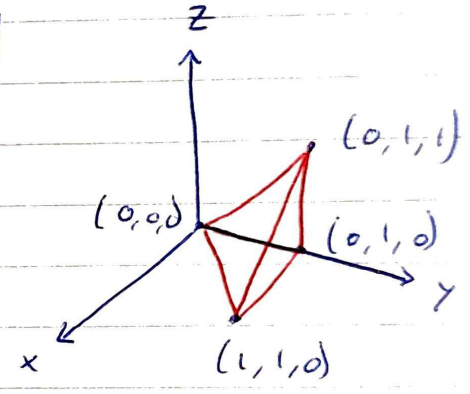
$$= \int_0^1 \left[3xy^2 + 3x^2y^2 + 2xy^3\right]_{y=0}^{y=\sqrt{x}}\,dx$$

$$= \int_0^1 3x^2 + 3x^3 + 2x^{\frac{5}{2}}\,dx$$

$$= \left[x^3 + \frac{3}{4}x^4 + \frac{4}{7}x^{\frac{7}{2}}\right]_{x=0}^{x=1}$$

$$= \frac{65}{28}$$

5) $\iiint_E xz\,dV$



Bottom of tetrahedron:
$0 \le y \le 1, \quad 0 \le x \le y$

Top of tetrahedron:
$z = y - x$

$\Rightarrow$ Inside tetrahedron:
$0 \le z \le y - x$

$$= \int_0^1 \int_0^y \int_0^{y-x} xz\,dz\,dx\,dy$$

$$= \int_0^1 \int_0^y \left[\frac{1}{2}xz^2\right]_{z=0}^{z=y-x}\,dx\,dy$$

$$= \int_0^1 \int_0^y \frac{1}{2}x^3 - x^2y + \frac{1}{2}xy^2\,dx\,dy$$

$$= \int_0^1 \left[\frac{1}{8}x^4 - \frac{1}{3}x^3y + \frac{1}{4}x^2y^2\right]_{x=0}^{x=y}\,dy$$

$$= \int_0^1 \frac{1}{8}y^4 - \frac{1}{3}y^4 + \frac{1}{4}y^4\,dy$$

$$= \int_0^1 \frac{1}{24}y^4\,dy$$

$$= \left[\frac{1}{120}y^5\right]_0^1 = \frac{1}{120}$$

6) $\iiint_E x^2 \, dV$

$E: \; x^2 + y^2 \le 1, \quad z \ge 0, \quad z^2 \le 4x^2 + 4y^2$

In cylindrical coordinates:

$r^2 \le 1 \implies r \le 1$

$z^2 \le 4r^2 \implies z \le 2r$

$$= \int_0^{2\pi} \int_0^1 \int_0^{2r} r^3 \cos^2\theta \, dz \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^1 2r^4 \cos^2\theta \, dr \, d\theta$$

$$= \int_0^{2\pi} \left[\frac{2}{5} r^5 \cos^2\theta \right]_{r=0}^{r=1} d\theta$$

$$= \int_0^{2\pi} \frac{2}{5} \cos^2\theta \, d\theta$$

$$\cos^2\theta = \frac{1 + \cos 2\theta}{2}$$

$$= \frac{1}{5} \int_0^{2\pi} 1 + \cos 2\theta \, d\theta$$

$$= \frac{1}{5} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{2\pi} = \frac{2\pi}{5}$$

7) $\iiint_B x^2 + y^2 + z^2 \, dV$ $\qquad B = \{(x,y,z) : x^2 + y^2 + z^2 \le 1\}$

$$= \int_0^{\pi} \int_0^{2\pi} \int_0^1 \rho^2 \, \rho^2 \sin\phi \, d\rho \, d\theta \, d\phi$$

$$= \int_0^{\pi} \int_0^{2\pi} \left[\frac{1}{5} \rho^5 \sin\phi \right]_{\rho=0}^{\rho=1} d\theta \, d\phi$$

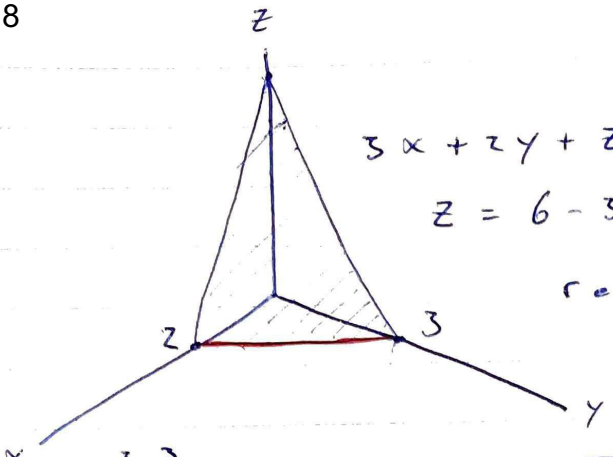
$$= \int_0^{\pi} \int_0^{2\pi} \frac{1}{5} \sin\phi \, d\theta \, d\phi$$

$$= \int_0^{\pi} \frac{2\pi}{5} \sin\phi \, d\phi$$

$$= \frac{2\pi}{5} \left[-\cos\phi \right]_0^{\pi}$$

$$= \frac{4\pi}{5}$$

Question 8



$3x + 2y + z = 6$

$z = 6 - 3x - 2y$

red line: $3x + 2y = 6$

$\Rightarrow y = 3 - \frac{3}{2}x$

$$A = \int_0^2 \int_0^{3-\frac{3}{2}x} \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\, dy\, dx$$

$$= \int_0^2 \int_0^{3-\frac{3}{2}x} \sqrt{1 + (-3)^2 + (-2)^2}\, dy\, dx$$

$$= \int_0^2 \int_0^{3-\frac{3}{2}x} \sqrt{14}\, dy\, dx$$

$$= \int_0^2 3\sqrt{14} - \frac{3}{2}\sqrt{14}\, x\, dx$$

$$= \left[3\sqrt{14}\, x - \frac{3}{4}\sqrt{14}\, x^2\right]_0^2 = 3\sqrt{14}$$

Question 9

$$\iiint_E \sqrt{x^2+y^2}\, dV =$$

$$\int_0^{2\pi} \int_0^4 \int_{-5}^4 r^2\, dz\, dr\, d\theta =$$

$$\int_0^{2\pi} \int_0^4 9r^2\, dr\, d\theta =$$

$$\int_0^{2\pi} \left[3r^3\right]_0^4 d\theta =$$

$$\int_0^{2\pi} 192\, d\theta = 384\pi$$

E lies inside cylinder $x^2 + y^2 = 16$ (radius 4) and between $z = -5$ and $z = 4$

Question 10

[redacted] $\iiint z \, dV =$

$\int_0^{\pi/2} \int_0^{\pi/2} \int_1^2 \rho \cos\phi \, \rho^2 \sin\phi \, d\rho \, d\theta \, d\phi =$

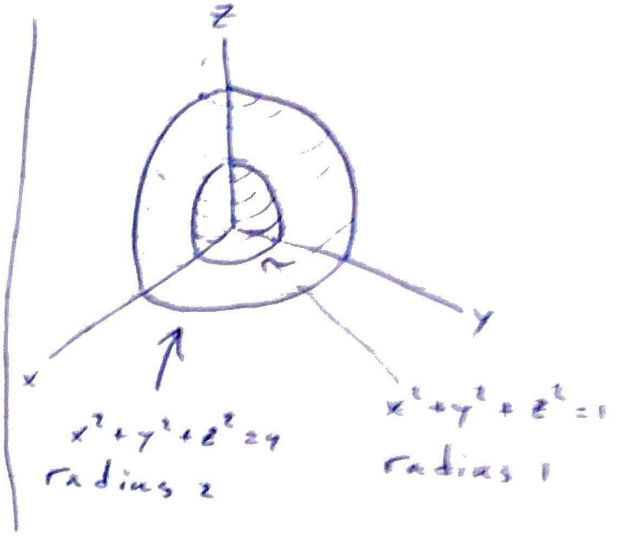
$\int_0^{\pi/2} \int_0^{\pi/2} \cos\phi \sin\phi \left[\frac{\rho^4}{4}\right]_1^2 d\theta \, d\phi =$

$\int_0^{\pi/2} \int_0^{\pi/2} \frac{15}{4} \cos\phi \sin\phi \, d\theta \, d\phi =$

$\int_0^{\pi/2} \frac{15}{8}\pi \cos\phi \sin\phi \, d\phi =$

$\int_0^{\pi/2} \frac{15\pi}{16} \sin 2\phi \, d\phi =$

$\frac{15\pi}{32} \left[-\cos 2\phi\right]_0^{\frac{\pi}{2}} = \frac{15\pi}{16}$



$\sin 2\phi = 2\sin\phi\cos\phi$

Question 11

[redacted]

$x = uv, \quad y = vw, \quad z = uw$

$\det\begin{pmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{pmatrix} = \det\begin{pmatrix} v & u & 0 \\ 0 & w & v \\ w & 0 & u \end{pmatrix}$

$= v \cdot wu - u \cdot -wv = 2uvw$

Question 12

[redacted] $\iint_R x^2 \, dx \, dy =$

$\iint 4u^2 \cdot 6 \, du \, dv =$

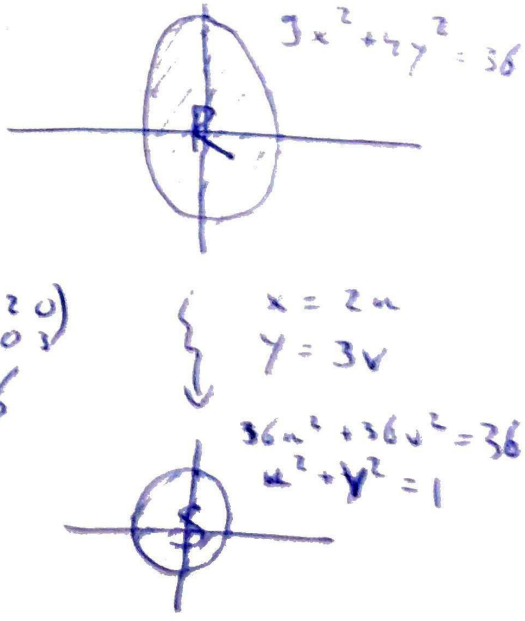
$\int_0^{\pi/2} \int_0^2 24 r^3 \cos^2\theta \, dr \, d\theta =$

$\int_0^{2\pi} \cos^2\theta \left[6r^4\right]_0^1 d\theta =$

$6\int_0^{2\pi} \cos^2\theta \, d\theta = 3\int_0^{2\pi} 1 + \cos 2\theta \, d\theta$

$= 3\left[\theta + \frac{1}{2}\sin 2\theta\right]_0^{2\pi} = 6\pi$

$\frac{\partial(x,y)}{\partial(u,v)} = \det\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} = 6$



Question 13

[redacted] $f(x,y,z) = \sqrt{x^2+y^2+z^2}$

$$\nabla f(x,y,z) = \left(\frac{1}{2\sqrt{x^2+y^2+z^2}}\cdot 2x,\ \frac{1}{2\sqrt{x^2+y^2+z^2}}\cdot 2y,\ \frac{1}{2\sqrt{x^2+y^2+z^2}}\cdot 2z\right)$$

$$= \frac{1}{\sqrt{x^2+y^2+z^2}}\,(x,\ y,\ z)$$

ii) $f(x,y) = \ln(x+2y)$

$$\nabla f(x,y) = \left(\frac{1}{x+2y},\ \frac{2}{x+2y}\right) = \frac{1}{x+2y}\,(1,\ 2)$$

Question 14

[redacted] $\int_C y \sin z \, ds =$

$C: t \mapsto (\cos t, \sin t, t)$, $t \in [0, 2\pi]$

$$\int_0^{2\pi} \sin^2 t \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2}\, dt$$

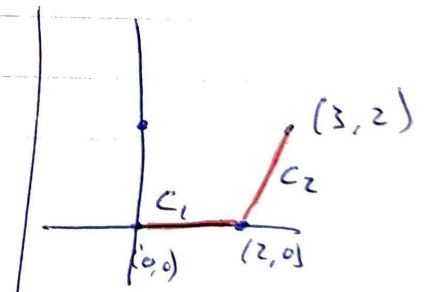
$$= \int_0^{2\pi} \sqrt{2}\, \sin^2 t \, dt$$

$\sin^2 t = \frac{1-\cos 2t}{2}$

$$= \frac{\sqrt{2}}{2}\int_0^{2\pi} 1 - \cos 2t \, dt$$

$$= \frac{\sqrt{2}}{2}\left[t - \frac{1}{2}\sin 2t\right]_0^{2\pi} = \pi\sqrt{2}$$

ii) $\int_C xy\,dx + (x-y)\,dy =$



$$\underbrace{\int_{C_1} xy\,dx}_{=0 \text{ because } y=0} + \int_{C_2} xy\,dx + \underbrace{\int_{C_1} (x-y)\,dy}_{=0 \text{ because } \frac{\partial y}{\partial t}=0} + \int_{C_2} (x-y)\,dy =$$

$C_1: t \mapsto (t, 0)$, $t \in [0, 2]$

$C_2: t \mapsto (2+t, 2t)$, $t \in [0, 1]$

$$\int_0^1 (4t + 2t^2)\frac{\partial x}{\partial t}\,dt + \int_0^1 (2-t)\frac{\partial y}{\partial t}\,dt =$$

$$\int_0^1 4t + 2t^2\,dt + \int_0^1 4 - 2t\,dt =$$

$$\left[2t^2 + \frac{2}{3}t^3\right]_0^1 + \left[4t - t^2\right]_0^1 = \frac{17}{3}$$